



Community Hero - Hyperlocal Problem Solver

Background

Communities frequently face issues such as potholes, water leakages, damaged streetlights, waste management concerns, and public infrastructure challenges. Reporting these issues is often fragmented, difficult to track, and lacks transparency.

Challenge

Build a platform that enables citizens to identify, report, validate, track, and resolve community issues through collaboration, data, and intelligent automation. The solution should encourage transparency, accountability, and community participation.

Example Features:

- Image and video-based issue reporting
- AI-powered issue categorization
- Geo-location and mapping
- Community verification
- Real-time issue tracking
- Impact dashboards
- Predictive insights
- Gamification for citizen engagement

Evaluation Focus

The solution should demonstrate how AI can help communities address local challenges more efficiently through improved reporting, verification, tracking, and resolution of issues.

Submission: JanSetu

1. Solution Overview

JanSetu shifts the civic grievance paradigm from standard 'passive complaints lists' to an **active, self-triage, and self-escalating operating system**. Instead of forcing citizens to navigate confusing municipal portals, select exact categories, or manually search for geolocations, JanSetu exposes a **Zero-UI voice interface** where citizens speak naturally in English, Hindi, or mixed dialects.

By utilizing an orchestration of specialized Gemini-powered agents, the system automates what was previously a friction-heavy manual administrative loop. Citizens reporting an issue get a structured ticket logged within 10 seconds. The ticket calculates dynamic priority scores, geo-clusters adjacent reports to prevent duplicate database bloat, and monitors SLAs autonomously. If municipal services fail to respond, JanSetu drafts and publishes formal escalation letters to municipal heads.

JanSetu targets the core of community engagement by closing the feedback loop. By verifying resolutions using comparative computer vision, the platform prevents municipal fraud and ensures real physical progress is recorded publicly on the audit trail.

2. Multi-Agent Pipeline & Deep Depth

The core innovation of JanSetu is its directed multi-agent loop, dividing tasks among independent agents to deliver deep agentic reasoning:

Agent	Core Reasoning & Job Description	API / Model
Triage Agent	Translates raw speech, categorizes context, extracts landmarks, and geocodes coords.	Gemini 1.5 Pro
Duplicate Detector	Groups spatial-temporal overlaps to prevent ticket duplication and spam.	Gemini 1.5 Pro
COI Engine	Formulates dynamic priority score based on hazard risk, density, and infrastructure type.	Gemini 1.5 Pro
SLA Watchdog	Queries live web for nodal government officials and drafts formal escalation letters.	Gemini + Search Grounding
Verification Agent	Compares before/after resolution photos using visual reasoning to prevent municipal fraud.	Gemini Vision

Each agent runs asynchronously on Firebase serverless pipelines, keeping the system decoupled and protecting API secret keys on the backend rather than in the client browser.

3. Key Features & Product Experience

- **Zero-UI Voice Reporting:** Transcribes and structures mixed-dialect voice notes in seconds.
- **Google Lens-Style Drag-and-Drop:** Clean custom drag-over dropzone with visual feedback and remove states.
- **COI-Ranked Dashboard:** Ranks tasks by Cost of Inaction score instead of standard chronology.
- **Heatmaps & Polygons:** Real-time Leaflet geocoding map tracking with automated risk-boundary clustering.
- **Grievance Directory:** Officer portal integrates real links for CPGRAMS, MyGov, MoHUA, Janaagraha, and Praja.
- **Public Ledger:** Immutable, transparent public audit log for accountability.

Visual Interface: Citizen View & Live Reports Feed

The screenshot displays the Jan Setu web application interface. The top navigation bar includes links for Home, Report Issue (highlighted), Live Map, Transparency Ledger, and Officer Portal. The main section is titled 'Report an Issue' with the subtitle 'Your voice matters. Describe the civic issue, and JanSetu will analyze, route, and escalate it automatically.'

The 'Report an Issue' form consists of three main sections:

- DESCRIBE THE PROBLEM:** A text input field with a placeholder example: 'e.g., Pipe bahar school ke paas toot gayi hai...'. Below the input is a small icon of a pipe with a drop.
- ATTACH PHOTO (OPTIONAL):** A dashed box for uploading a photo. It contains a camera icon and the text: 'Drag an image here or [upload a file](#). JPG, PNG, or WebP · Max 5 MB'.
- LOCATION:** A button labeled 'Use Current Location' with a location pin icon.

To the right of the form is the 'IN Live Citizen Reports' section, which features a 'INDIA-WIDE FEED' header. Below the header, it states: 'Recent civic issues automatically triaged, geocoded, and logged by JanSetu.'

The feed lists four recent reports:

- KAROL BAGH, NEW DELHI** (10 mins ago): "Water supply pipe leakage near Karol Bagh metro station, New Delhi. Water logging on main road." Category: **WATER LEAK**, Severity: **4/5**, Status: **OPEN**.
- SALT LAKE SECTOR V, KOLKATA** (25 mins ago): "Huge garbage pile lying near Salt Lake sector V bus stand, Kolkata. Breeding ground for mosquitoes." Category: **GARBAGE DUMP**, Severity: **3/5**, Status: **OPEN**.
- INDIRANAGAR, BENGALURU** (1 hour ago): "Potholes all over the flyover connecting Indiranagar to Domlur, Bengaluru. Highly dangerous for two wheelers." Category: **ROAD DAMAGE**, Severity: **5/5**, Status: **ESCALATED**.
- GACHIBOWLI, HYDERABAD** (3 hours ago): "Streetlight not working since last 3 weeks near Gachibowli flyover side lane, Hyderabad. Very dark at night." Category: **STREETLIGHT**, Severity: **2/5**, Status: **OPEN**.

4. Technologies & Google Integration

Technologies Used

Frontend Core: React 18, Vite (HMR bundling), and React Router DOM.

Design & Layout: Vanilla CSS using glassmorphism, responsive flex layouts, dynamic grid canvas background, and custom HSL color palettes.

Mapping: Leaflet.js map instance + HTML5 Geolocation API.

Google & Firebase Services Utilized

- **Gemini 1.5 Pro:** Handles core translation, NLP structures, priority scores, and letter generation.
- **Gemini Search Grounding:** Queries public records live to locate active officers and engineering contacts.
- **Gemini Vision:** Audits and compares original vs. resolution photos to verify resolved issues.
- **Firebase Firestore:** Real-time synchronization of issues database and transparency ledgers.
- **Firebase Authentication:** Citizen anonymous sessions & secure administrative officer portal credentials.
- **Firebase Storage & Hosting:** Hosts static assets under 5MB and serves production build at ``jansetu-dev.web.app``.